Supplementary Appendix for *Effort*

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This document is the appendix of the paper *Effort*, published in the NeurIPS 2026 Creative AI Track. It accompanies the release at `https://github.com/ahugh19/Effort`, and its section letters match the appendix sections of the paper.

A Model training and dataset

Effort uses Lodge [1], a two-stage coarse-to-fine diffusion network that generates dance motion conditioned on music, trained from scratch for this project. The trace analyzed here comes from the Global module, the coarse stage of the network. Music enters the model as 35-dimensional handcrafted audio features, and motion is represented as a 139-dimensional feature vector over 22 joints. The trace therefore records the optimization history of this specific run. The training data is FineDance [2], a motion-capture dataset of paired music and full-body 3D dance recordings performed by twenty-seven professional dancers, used in its standard training split. Training ran for 3,500 epochs on a single NVIDIA H100, about 51 hours of active GPU time. The released configuration records the optimizer settings: Adan with a learning rate of 4×10^{-4} held constant, batch size 128 with gradient accumulation of three for an effective batch of 384. We disabled gradient clipping, so the logged gradient norm is the raw norm. A logging callback writes one row per batch, giving 155,232 rows over 52,516 optimizer steps. Each row records the step, epoch, total and component losses, gradient-norm statistics, the learning rate, and binary indicators for loss spikes, plateaus, and exploding or vanishing gradients, and every indicator is false in every row.

The artwork uses three signals from the log: training loss, mean gradient norm, and gradient-norm volatility. Volatility is measured two ways. At the phase level, it is the standard deviation of the gradient norm over the phase's downsampled rows, and the ratio of that standard deviation to the phase mean drives the Effort mapping in Section C. At the per-batch level, the log stores a sign-change rate, the fraction of gradient coordinates whose sign flips between consecutive batches, and the sonifier in Section B uses it for rhythm.

A change-point detector divides the run into fourteen contiguous phases. The per-batch log is downsampled to every fifteenth row in acquisition order, and the PELT algorithm from the `ruptures` library is run on the smoothed loss and on the gradient norm with a radial-basis-function cost, a penalty of 10, a merge window of 200 rows, and a minimum segment length of 500 rows, all in downsampled rows. The two sets of breakpoints are merged and the minimum-length constraint is enforced, which yields fourteen phases. Table 2 lists the per-phase statistics. The run opens with a phase of large, unstable gradients, followed by thirteen lower-amplitude phases of gradual convergence in which the mean loss falls from about 0.116 to about 0.0098. The artistic transformations described below operate on this continuous training history.

*Human dance performance and improvisation by Qianni Wang.

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B Translating the training trace into music

The first translation turns the per-batch training log into a musical score through an artist-designed, rule-based procedure. Given the same log and settings, the procedure regenerates the same symbolic score. The score is written for a mechanical player piano with a small chamber ensemble, in D natural minor, in 4/4, at 70 beats per minute. The player piano carries the melody, strings and a low synthesizer drone support it, and celesta, harp, contrabass, and string pad enter occasionally. Four quantities from the log control four musical parameters, each normalized against this run's own range.

1. **Register.** The gradient norm sets the register of the melody and accompaniment. Larger gradients move the material upward.
2. **Rhythmic density.** The gradient sign-change rate sets the number and spacing of note onsets. Stable intervals give sparse, sustained notes, and volatile intervals give denser subdivisions and syncopation.
3. **Tremolo and chromatic inflection.** Bars with a change in loss above the 82nd percentile across the piece receive a string tremolo and unresolved chromatic neighbor tones. The lines are otherwise diatonic.
4. **Spectral brightness.** The gradient level sets the brightness of the synthesizer pad, so the pad darkens after the high-gradient opening.

Data controls register, rhythmic density, the accents, and the phase durations. The key, the harmonic progressions, the instrumentation, and the five-part form are artist-designed.

The fourteen training phases are laid out in training order on one linear clock. Each phase receives bars in proportion to its length, with a floor of two bars to keep every phase audible. Music time therefore follows training time at bar-level resolution. Over this clock the instrumentation follows a fixed five-part shape, a piano-led opening, a build, a short caesura, a denser passage, and a decay as the layers drop out. The score ends on an open fifth, D and A, and its final four bars slow from 70 to about 38 beats per minute. The Effort words in Section C are indexed by the same fourteen phases, so the two translations share one clock.

C Translating the training trace into Laban Effort words

The second translation gives one Laban Effort word to each of the fourteen phases through the three Effort factors of *Weight*, *Time*, and *Space* [3]. The mapping is an artist-designed convention applied to the statistics of this run.

For phase p, let μ_p be its mean gradient norm, σ_p the standard deviation of the gradient norm, and τ_p its temporal trend, the slope of a least-squares fit of the gradient norm over the phase. Let m_μ and m_τ be the medians of μ_p and $|\tau_p|$ over the fourteen phases. *Weight* is strong when $\mu_p \geq m_\mu$ and light otherwise. *Time* is sudden when $\sigma_p \geq \mu_p$ and sustained otherwise. *Space* is direct when $|\tau_p| \geq m_\tau$ and indirect otherwise. *Weight* and *Space* compare a phase with the whole run, and *Time* compares a phase's variability with its own mean. The three binary factors give the eight Effort words in Table 1.

Weight	Time	Space	Effort
light	sustained	indirect	*Float*
light	sustained	direct	*Glide*
light	sudden	direct	*Dab*
light	sudden	indirect	*Flick*
strong	sustained	direct	*Press*
strong	sustained	indirect	*Wring*
strong	sudden	direct	*Punch*
strong	sudden	indirect	*Slash*

Table 1: Artist-designed mapping from the three binary Laban Effort factors to eight Effort words. Each training phase is assigned one combination.

The first phase, with large and unstable gradients and a steep downward trend, is strong, sudden, and direct, the Effort *Punch*. It is the only sudden phase, since every later phase has a standard

Phase	Start step	Epoch	μ_p	σ_p	τ_p	Weight	Time	Space	Effort
1	0	0	2.741	5.017	-1.6×10^{-2}	strong	sudden	direct	*Punch*
2	2593	172	0.343	0.273	-2.3×10^{-4}	strong	sustained	direct	*Press*
3	5610	374	0.235	0.175	$+9.4 \times 10^{-6}$	light	sustained	indirect	*Float*
4	8704	580	0.233	0.200	-9.8×10^{-5}	light	sustained	direct	*Glide*
5	11260	750	0.248	0.198	$+1.0 \times 10^{-4}$	light	sustained	direct	*Glide*
6	15019	1001	0.272	0.170	$+3.2 \times 10^{-5}$	light	sustained	indirect	*Float*
7	18292	1219	0.283	0.180	-1.4×10^{-5}	light	sustained	indirect	*Float*
8	21155	1410	0.313	0.245	$+1.3 \times 10^{-4}$	light	sustained	direct	*Glide*
9	25016	1667	0.341	0.204	$+5.3 \times 10^{-5}$	light	sustained	direct	*Glide*
10	28902	1926	0.370	0.218	$+6.5 \times 10^{-5}$	strong	sustained	direct	*Press*
11	33530	2235	0.386	0.221	$+4.6 \times 10^{-5}$	strong	sustained	indirect	*Wring*
12	36394	2426	0.398	0.247	$+2.2 \times 10^{-5}$	strong	sustained	indirect	*Wring*
13	43374	2891	0.379	0.209	$+1.3 \times 10^{-5}$	strong	sustained	indirect	*Wring*
14	45931	3062	0.370	0.207	$+3.2 \times 10^{-6}$	strong	sustained	indirect	*Wring*

Table 2: The fourteen phases of the training run. Start step is the first logged optimizer step of each phase. Epoch is the approximate epoch at that boundary. The phases are contiguous, so a phase runs to the next phase's start. The remaining columns give the mean gradient norm μ_p, the gradient-norm standard deviation σ_p, the temporal trend τ_p, the three binary factors, and the resulting Effort word. The median of μ_p is 0.342 and the median of $|\tau_p|$ is 4.9×10^{-5}.

deviation below its mean. The run then passes through *Press*, settles into *Float* and *Glide* through its low-gradient middle, and ends in *Press* and *Wring* as the gradient rises again. Table 2 lists the full assignment. The resulting Effort sequence is a choreographic interpretation constructed through our selection of signals, segmentation, thresholds, and factor mapping. In the performance the dancer receives each word as an open score. The word sets the quality of a movement, and the pose and trajectory remain their choice.

D Suno re-performance and temporal alignment

The rule-based score fixes the melody, harmony, form, phase durations, and tempo. For the video we passed its rendering to Suno [4], a generative music service, in its Cover mode. This step is a generative re-performance outside the deterministic mapping, and Suno can return a different cover on each request, so we keep the selected outputs as fixed artifacts.

The presented cover was generated with model V5.5 as an instrumental, with Audio Influence 85%, Style Influence 15%, and Weirdness 5%, from the following prompt. We generated six candidates with the same prompt and settings and selected the take closest to the source player-piano motif and its restrained dynamic. The file is used as Suno delivered it.

> COVER FAITHFULLY — preserve the uploaded melody, chord changes, tempo (keep the final ritardando), and section structure. Keep the dynamics restrained from start to end: quiet, chamber-scale, mezzo-piano at most. The "climax" is only a slight thickening of texture, then it thins again. A small, cold ensemble: a mechanical player piano stating the melody with precise, unhurried articulation; one solo cello answering low and dry; a faint, even eighth-note pulse in muted low strings, felt more than heard; sparse high violin harmonics for air. Wide space between phrases, generous silence, cool and exact intonation. D minor, approximately 70 BPM, 4/4. The ending slows and settles on a bare open fifth, unresolved, fading into a dry, cold room tail. Mood: wintry, elegant, austere, lucid, still; mechanical calm rather than passion. Instrumental only, close-miked modern classical recording, high detail, quiet mastering.

The cover changes the timing of the score, so we align the visual materials to it with chroma-based dynamic time warping, a monotone map from score time to cover time. The plotted curves, bar and phase boundaries, and Effort-word changes all follow the timing of the audio in the video.

The dance is conditioned on a second cover of the same score, arranged in the style of Kunqu opera accompaniment. The model produces more coherent motion from this cover than from the sparse texture of the presented cover, and its movement carries the stillness and austerity closest to the finished work. This cover keeps the key, phrase structure, and five-part form of the score at close

to the same tempo, and differs from the presented cover in instrumentation and surface character. The generated dance is aligned to the presented cover with the same warping. It was generated with model V4.5, Audio Influence 85%, and Weirdness 0%, from the following prompt.

> COVER, REARRANGED IN THE STYLE OF KUNQU OPERA ACCOMPANIMENT. Keep the source's slow tempo, minor-key character, and melodic contour. Replace the Western orchestration with traditional Kunqu instrumentation. D minor with pentatonic flavor, approximately 65 BPM, 4/4. Production: bamboo flute (qudi or dizi) carrying the melodic line, sheng (mouth organ) sustained drones, pipa or sanxian plucked accents on phrase boundaries, soft guzheng arpeggios, light wooden percussion (wooden clappers and paigu on downbeats), small bells. Subtle wordless female vocalise in Kunqu falsetto style — breathy, soft, ornamented — hovering as background, sparse, only in climax and outro. Natural temple-hall reverb. Mood: meditative, classical Chinese, sacred lament, slow and flowing. Suitable as score for Kunqu and classical Chinese stage choreography. Mostly instrumental.

E Reproducibility and retained artifacts

The release at `https://github.com/ahugh19/Effort` contains the per-batch training log, the phase summary, the symbolic score and its mastered render, the two Suno covers, the alignment timeline, and the code for each deterministic step. The pipeline runs in the following order. Steps 1 and 5 are outside the release, and their outputs are kept as fixed files.

1. Train Lodge on FineDance and write the per-batch log.
2. Segment the log into fourteen phases.
3. Generate the symbolic score and, optionally, a dry preview render.
4. Assign one Effort word to each phase.
5. Generate and select the Suno cover.
6. Align the score to the presented cover by dynamic time warping.
7. Render the curve, word, dancer, and skeleton layers on the aligned timeline.

The exact parameters of steps 2, 3, 4, and 6 are recorded in the released scripts. The mastered render stays a fixed file because mastering happened outside the package, and the covers stay fixed files because the service returns a different result on each request. The FineDance recordings remain with their authors [2] under the dataset license, and the released material is provided under the license stated in the release README.

References

[1] Ronghui Li, Yuxiang Zhang, Yachao Zhang, Hongwen Zhang, Jie Guo, Yan Zhang, Yebin Liu, and Xiu Li. Lodge: A coarse to fine diffusion network for long dance generation guided by the characteristic dance primitives. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[2] Ronghui Li, Junfan Zhao, Yachao Zhang, Mingyang Su, Zeping Ren, Han Zhang, Yansong Tang, and Xiu Li. Finedance: A fine-grained choreography dataset for 3d full body dance generation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 10234–10243, 2023.

[3] Rudolf Laban and F. C. Lawrence. *Effort*. Macdonald and Evans, London, 1947.

[4] Suno, Inc. Suno. Generative music service, https://suno.com, 2026. Accessed 2026.